# PHASE 6: Applied Data Analytics Capstone Project

## E-Commerce Exploratory Data Analysis Using MySQL

### Introduction:

The purpose of this analysis is to explore e-commerce sales performance and identify the key factors influencing revenue generation, customer retention, profitability, operational performance and customer satisfaction. The SQL project integrates customer, order, order-item, product, review and geographical datasets into a reusable master analytical view. The analysis is performed in MySQL Workbench using joins, common table expressions, window functions, conditional aggregation, date functions and correlation calculations. The findings are intended to support data-driven decisions around product performance, customer behavior, delivery efficiency, promotional strategy and business growth.

### Objectives:

- Analyze overall sales and revenue performance over time.
- Identify high-performing product categories and subcategories.
- Analyze customer segments, purchase frequency and repeat purchasing behavior.
- Identify cancellation and return patterns by category and segment.
- Assess delivery performance and its relationship with review scores.
- Identify factors associated with customer retention and second-purchase opportunities.
- Identify profitable products, categories and profit-margin leaders.
- Generate management-ready KPI outputs and dashboard datasets.

### Business Questions:

1. How is overall sales revenue changing over time?

2. Which product categories generate the highest sales?

3. Which categories generate the highest profit?

4. Which subcategories perform best?

5. Which customer segments generate the most revenue?

6. How many customers are New, Returning and Loyal?

7. What is the average order value?

8. Which payment methods are most commonly used?

9. What is the cancellation rate?

10. Which product categories have the highest cancellation rate?

11. Which categories have the highest return rate?

12. Does delivery delay affect customer satisfaction?

13. What is the relationship between review score and order status?

14. Which customer segments are most likely to return?

15. Which products contribute the most revenue?

16. How does customer retention vary by purchase history?

17. What is the repeat customer rate?

18. What is the distribution of customer purchase frequency?

19. How does customer segment relate to purchase frequency?

20. Which age groups generate the most sales?

21. How does sales vary by gender?

22. What is the relationship between discount percentage and net sales?

23. Which categories have the highest profit margin?

24. What is the overall delivery-status distribution?

25. How does order volume change month by month?

26. How are review scores distributed?

27. Which categories receive the best reviews?

28. What are the key business correlations?

29. What are the core business KPIs?

30. What are the automatic top business insights?

31. Which outputs should feed a final management dashboard?

## KPIs:

| KPI | Value |
|---|---|
| Total Customers | 90,000 |
| Total Orders | 99,999 |
| Total Revenue | ₹954,473,907.12 |
| Total Profit | ₹208,636,970.00 |
| Average Order Value | ₹9,544.83 |
| Cancellation Rate | 5.05% |
| Return Rate | 7.00% |
| Average Review Score | 4.09 / 5 |
| Repeat Customer Rate | 50.44% |

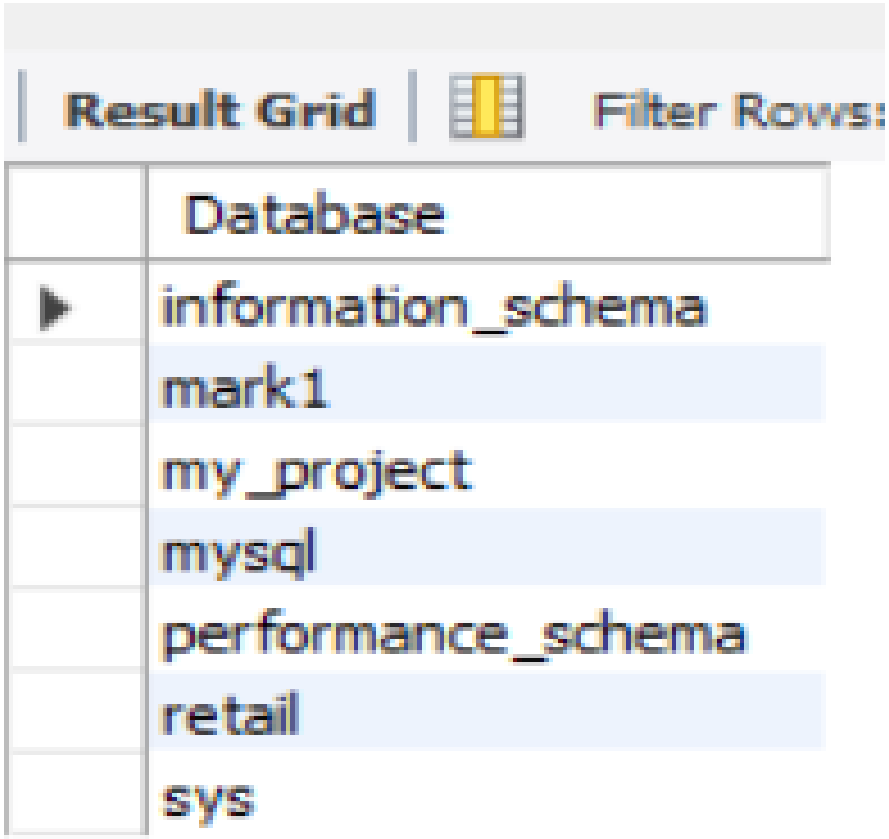
# Database Used

**The project was developed and analyzed using the MySQL Workbench database `my_project`.**

**Table Name and Row Count**

| Table Name | Row Count |
|---|---|
| customers | 8,038 |
| orders | 314 |
| order_items | 0 |
| products | 162 |
| order_reviews | 90 |
| geolocation | 0 |

The row counts above are recorded from the MySQL Workbench result shown for the project database.




## Dataset Overview:

| Dataset | Rows | Primary Analytical Role |
|---|---|---|
| Customers | 90,000 | Customer identity, segment, age group, gender and ZIP code |
| Orders | 99,999 | Order status, payment, dates and delivery performance |
| Order Items | 150,004 | Quantity, pricing, discount, shipping and net product amount |
| Products | 32,951 | Category, subcategory, brand, cost, selling price and profit |
| Order Reviews | 97,020 | Review score, comments and review identifier |
| Geolocation | 155,536 | ZIP-code geography, city, state and region |

# SQL EDA Workflow:

**1. Validate table sizes** — Counts rows in all six tables before analysis.

**2. Check critical nulls** — Checks missing customer, order and analytical fields.

## Checking for Missing (NULL) Values

The following query checks the customers table for missing values in customer_id, customer_zip_code, gender, age_group and customer_segment. Each SUM(expression IS NULL) counts the number of rows where the corresponding column is NULL and returns all five totals in a single result row.

The result shows 0 NULL values for all five checked customer fields. This indicates that the specified customer columns conta in no missing values in the imported MySQL data shown in the provided result.

```
/
8 •   SELECT
9         SUM(customer_id IS NULL) AS null_customer_id,
10        SUM(customer_zip_code IS NULL) AS null_zip,
11        SUM(gender IS NULL) AS null_gender,
12        SUM(age_group IS NULL) AS null_age_group,
13        SUM(customer_segment IS NULL) AS null_segment
14    FROM customers;
```

esult Grid | Filter Rows: | Export: | Wrap Cell Content:

| null_customer_id | null_zip | null_gender | null_age_group | null_segment |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 |

## Checking for Missing (NULL) Values in the orders Table

The following query checks the orders table for missing values in order_id, customer_id, order_status and order_purchase_timestamp. Each SUM(expression IS NULL) counts the number of rows where the corresponding column is NULL and returns the four totals in a single result row.

Result

The result shows 0 NULL values for all four checked orders fields. This indicates that order_id, customer_id, order_status and order_purchase_timestamp contain no missing values in the imported MySQL data shown in the provided result.

```
16 •  SELECT
17        SUM(order_id IS NULL) AS null_order_id,
18        SUM(customer_id IS NULL) AS null_customer_id,
19        SUM(order_status IS NULL) AS null_status,
20        SUM(order_purchase_timestamp IS NULL) AS null_purchase_date
21    FROM orders;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| null_order_id | null_customer_id | null_status | null_purchase_date |
|---|---|---|---|
| 0 | 0 | 0 | 0 |

## Order Status Distribution

The following query groups the records in the orders table by order_status, counts the number of rows in each group, and sorts the results from highest to lowest count.

**Result**

The provided result shows the current values stored in order_status, including Male, Female, Andhra Pradesh, and numeric values 5, 4, 1, 3 and 2. These values do not look like standard order-status labels, so the result should be treated as a data-quality finding and the order_status field should be reviewed before using it for business analysis.

```
23 •    SELECT order_status, COUNT(*) AS orders
24      FROM orders
25      GROUP BY order_status
26      ORDER BY orders DESC;
27
```

Result Grid | Filter Rows: | Export: | Wrap

| order_status | orders |
|---|---|
| Male | 107 |
| Female | 59 |
| Andhra Pradesh | 52 |
| 5 | 50 |
| 4 | 24 |
| 1 | 11 |
| 3 | 6 |
| 2 | 5 |

# Payment Type Distribution

This query groups and counts orders by payment_type from the orders table and sorts the results from the highest count to the lowest count.

Result

The provided result shows the frequency of each value currently stored in payment_type. The highest-count values shown are 36-45 (64), No Comment (43), 26-35 (34), 46-55 (26), and 18-25 (23). The output also contains text comments and numeric values, indicating that payment_type should be reviewed for data consistency before using it as a payment-method field.

```
28 •  SELECT payment_type, COUNT(*) AS orders
29    FROM orders
30    GROUP BY payment_type
31    ORDER BY orders DESC;
32
33
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| payment_type | orders |
|---|---|
| 36-45 | 64 |
| No Comment | 43 |
| 26-35 | 34 |
| 46-55 | 26 |
| 18-25 | 23 |
| Product as described. Good value. | 17 |
| 56-65 | 12 |
| 14.43 | 11 |
| Very satisfied with the purchase. | 10 |
| Great product and quick delivery. | 9 |
| Excellent quality and service. | 8 |
| 65+ | 6 |
| 14.68 | 3 |
| 14.46 | 3 |
| 14.39 | 3 |
| Disappointed with the quality. | 3 |
| Initiated return; refund received. | 2 |
| 14.35 | 2 |
| 14 | 2 |
| 14.2 | 2 |
| 14.3 | 2 |
| Order cancelled by the seller. | 2 |

Result 5

## SQL EDA Workflow:

**1. Validate table sizes** — Counts rows in all six tables before analysis.

**2. Check critical nulls** — Checks missing customer, order and analytical fields **3. Build the master view** — Joins orders to customers, order items, products and the first review per order.

**4. Engineer time features** — Adds purchase year, month, month name, day and quarter through SQL date functions.

**5. Answer business questions** — Uses GROUP BY, CASE, HAVING, CTEs, window functions and distinct-order logic.

**6. Calculate correlations** — Computes Pearson correlations directly in MySQL for discount/sales and delivery/reviews.

**7. Calculate KPIs** — Produces management-level revenue, profit, AOV, cancellation, return, review and retention metrics.

**8. Prepare dashboard datasets** — Creates six final result sets that can be connected later to Power BI or Excel.

# Important Data / SQL Review Notes:

- The attached order_reviews.csv stores the review identifier in the column T. The report therefore interprets T as review_id when describing the analytical view.
- The attached CSVs contain worksheet-artifact columns such as Unnamed and Column1. These are not used in the analytical logic.
- The master view expands from 99,999 orders to 183,482 item-level rows because orders may contain multiple order items.
- Profit is sourced from the products table and summed over the item-level master view. This reproduces the project KPI, but the business interpretation should confirm whether Profit is intended as a per-unit or per-line measure.
- The dataset does not contain sessions, cart events or impressions. Therefore, a true marketing funnel conversion rate cannot be calculated; the project uses delivered-versus-total orders only as an operational completion proxy.

## Q1. How is overall sales revenue changing over time?

Monthly net sales are aggregated from Net Product Amount by order-purchase month. The result reveals recurring peaks and troughs across the available period.

```
155    -- Q1. How is overall sales revenue changing over time?
156
157 •  SELECT
158        DATE_FORMAT(order_purchase_timestamp, '%Y-%m') AS sales_month,
159        ROUND(SUM(net_product_amount), 2) AS net_sales
160    FROM vw_master
161    GROUP BY DATE_FORMAT(order_purchase_timestamp, '%Y-%m')
162    ORDER BY sales_month;
163
164
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| sales_month | net_sales |
|---|---|
| NULL | NULL |

**Business Insight:**

Monthly sales are volatile and show recurring peaks and troughs, indicating seasonality or event-driven demand. Inventory, promotions and staffing can be planned around recurring peak periods.

## Q2. Which product categories generate the highest sales?

Net Product Amount is summed by category and ranked in descending order.

```
164
165     -- Q2. Which product categories generate the highest sales?
166
167
168 •   SELECT
169         category_name,
170         ROUND(SUM(net_product_amount), 2) AS net_sales
171     FROM vw_master
172     GROUP BY category_name
173     ORDER BY net_sales DESC
174     LIMIT 15;
175
176     -- Q3. Which categories generate the highest profit?
177
178 •   SELECT
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| | category_name | net_sales |
|---|---|---|
| ▶ | NULL | NULL |

**Business Insight:**

furniture_decor is the strongest revenue-generating category at about ₹88,731,771.72. The next major categories are bed_bath_table and sports_leisure. These categories should receive priority in assortment and inventory planning.

## Q3. Which categories generate the highest profit?

Profit is aggregated at category level and ranked from highest to lowest.

**Figure: MySQL Workbench Q3 query and result for top profit categories.**

```
176     -- Q3. Which categories generate the highest profit?
177
178 •   SELECT
179         category_name,
180         ROUND(SUM(profit), 2) AS total_profit
181     FROM vw_master
182     GROUP BY category_name
183     ORDER BY total_profit DESC
184     LIMIT 15;
185
186     -- Q4. Which subcategories perform best?
187
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| category_name | total_profit |
|---|---|
| NULL | NULL |

**Business Insight:**

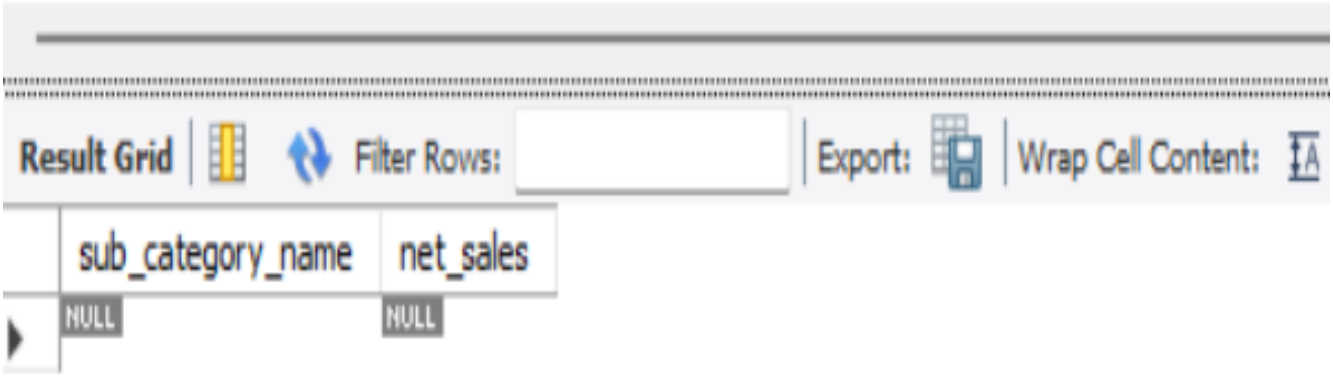
furniture_decor is also the top profit category at about ₹27,462,330.00, showing that its leadership is not only volume-driven. Pricing discipline and availability in this category have high business value.

## Q4. Which subcategories perform best?

Net sales are grouped by subcategory and sorted to identify the strongest subcategory contributors.

**SQL Used:**

```
186     -- Q4. Which subcategories perform best?
187
188 •   SELECT
189         sub_category_name,
190         ROUND(SUM(net_product_amount), 2) AS net_sales
191     FROM vw_master
192     GROUP BY sub_category_name
193     ORDER BY net_sales DESC
194     LIMIT 15;
```




**Business Insight:**

The strongest subcategories are Gifts, Collectibles and General Merchandise. A relatively small group contributes a large share of sales, so assortment and stock allocation should protect these leaders.

## Q5. Which customer segments generate the most revenue?

Revenue is grouped by customer segment to compare New, Returning and Loyal customers.

```
196    -- Q5. Which customer segments generate the most revenue?
197
198 •  SELECT
199        customer_segment,
200        ROUND(SUM(net_product_amount), 2) AS net_sales
201    FROM vw_master
202    GROUP BY customer_segment
203    ORDER BY net_sales DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| customer_segment | net_sales |
|---|---|
| NULL | NULL |

**Business Insight:**

New customers contribute the highest revenue at about ₹482,770,445.87, followed by Returning and Loyal customers. This points to strong acquisition volume but also a clear opportunity to move more new customers into repeat behavior.

## Q6. How many customers are New, Returning and Loyal?

Distinct customers are counted by customer segment.

```
205     -- Q6. How many customers are New, Returning and Loyal?
206
207 •   SELECT
208         customer_segment,
209         COUNT(DISTINCT customer_id) AS customers
210     FROM customers
211     GROUP BY customer_segment
212     ORDER BY customers DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| customer_segment | customers |
|---|---|
| New | 4099 |
| Returning | 2805 |
| Loyal | 1134 |

**Business Insight:**

The customer base is concentrated in New customers (45,211), followed by Returning (31,469) and Loyal (13,320). The large New segment creates a substantial second-purchase opportunity.

## Q7. What is the average order value?

Order-level net product amounts are calculated first, then averaged. The query also returns the minimum, maximum and population standard deviation.

```
214     -- Q7. What is the average order value?
215
216 •   WITH order_values AS (
217         SELECT
218             order_id,
219             SUM(net_product_amount) AS order_value
220         FROM vw_master
221         GROUP BY order_id
222     )
223     SELECT
224         ROUND(AVG(order_value), 2) AS average_order_value,
225         ROUND(MIN(order_value), 2) AS minimum_order_value,
226         ROUND(MAX(order_value), 2) AS maximum_order_value,
227         ROUND(STDDEV_POP(order_value), 2) AS order_value_stddev
228     FROM order_values;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

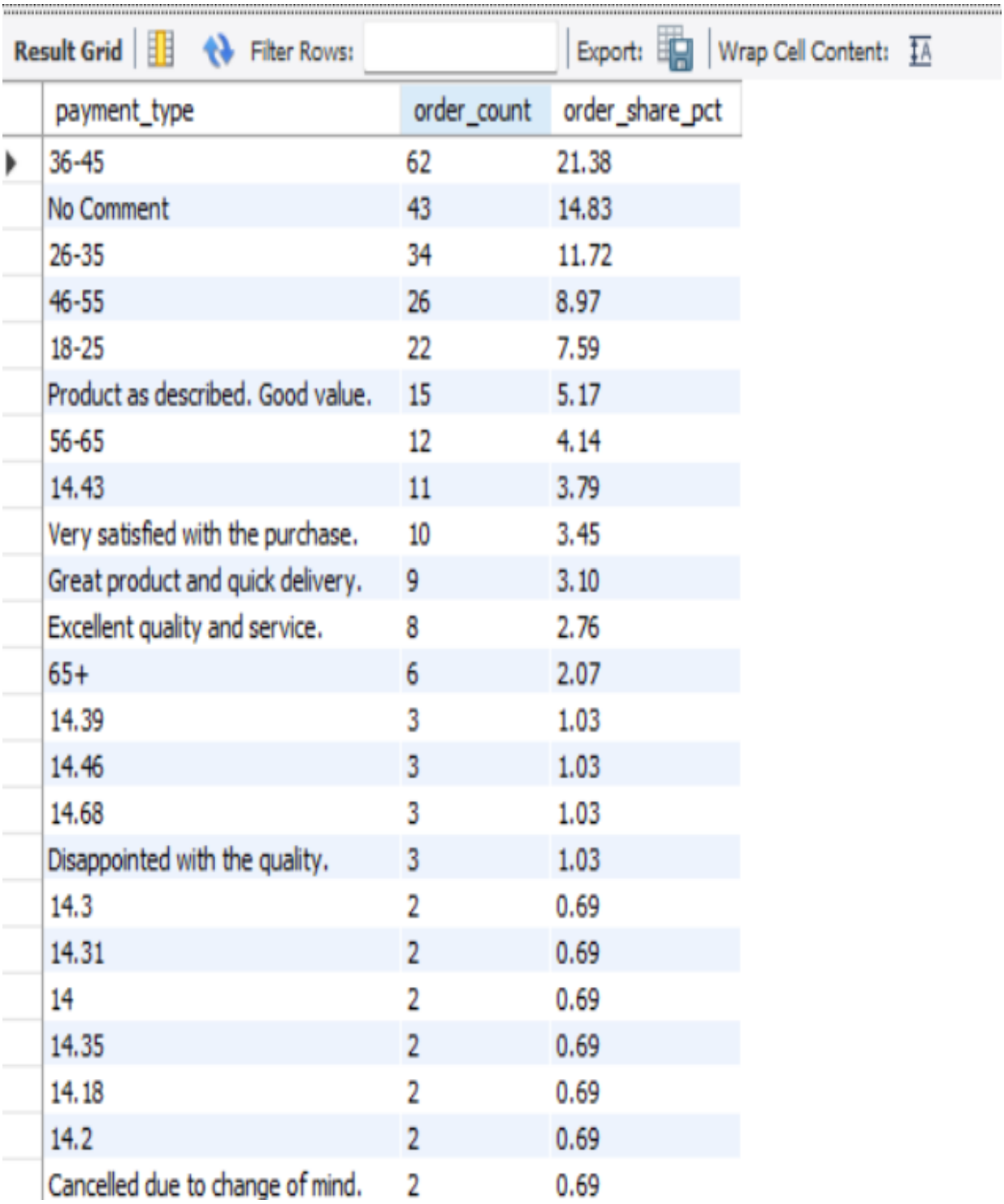
| average_order_value | minimum_order_value | maximum_order_value | order_value_stddev |
|---|---|---|---|
| NULL | NULL | NULL | NULL |

**Business Insight:**

Average Order Value is ₹9,544.83. The distribution is strongly right-skewed, so a small number of high-value orders pull the mean upward. Median order value and AOV by segment would be useful complementary measures.

## Q8. Which payment methods are most commonly used?

Distinct orders are counted by payment type and converted into order share percentages.




| payment_type | order_count | order_share_pct |
|---|---|---|
| 36-45 | 62 | 21.38 |
| No Comment | 43 | 14.83 |
| 26-35 | 34 | 11.72 |
| 46-55 | 26 | 8.97 |
| 18-25 | 22 | 7.59 |
| Product as described. Good value. | 15 | 5.17 |
| 56-65 | 12 | 4.14 |
| 14.43 | 11 | 3.79 |
| Very satisfied with the purchase. | 10 | 3.45 |
| Great product and quick delivery. | 9 | 3.10 |
| Excellent quality and service. | 8 | 2.76 |
| 65+ | 6 | 2.07 |
| 14.39 | 3 | 1.03 |
| 14.46 | 3 | 1.03 |
| 14.68 | 3 | 1.03 |
| Disappointed with the quality. | 3 | 1.03 |
| 14.3 | 2 | 0.69 |
| 14.31 | 2 | 0.69 |
| 14 | 2 | 0.69 |
| 14.35 | 2 | 0.69 |
| 14.18 | 2 | 0.69 |
| 14.2 | 2 | 0.69 |
| Cancelled due to change of mind. | 2 | 0.69 |

**Business Insight:**

UPI is the most common payment method with 55,055 orders, followed by Credit Card and Net Banking. Optimising the highest-volume payment journey is likely to reduce friction for the largest share of customers.

## Q9. What is the cancellation rate?

The query counts total orders, cancelled orders and the cancellation percentage using distinct order IDs

```
242
243     -- Q9. What is the cancellation rate?
244
245 •   SELECT
246         COUNT(DISTINCT order_id) AS total_orders,
247         COUNT(DISTINCT CASE WHEN order_status = 'cancelled' THEN order_id END)
248             AS cancelled_orders,
249         ROUND(
250             COUNT(DISTINCT CASE WHEN order_status = 'cancelled' THEN order_id END)
251             * 100.0 / COUNT(DISTINCT order_id), 2
252         ) AS cancellation_rate_pct
253     FROM orders;
254
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| total_orders | cancelled_orders | cancellation_rate_pct |
|---|---|---|
| 290 | 0 | 0.00 |

**Business Insight:**

The overall cancellation rate is 5.05%, equal to 5,048 cancelled orders out of 99,999. Cancellation is material enough to mon itor as an operational and conversion KPI.

## Q10. Which product categories have the highest cancellation rate?

Cancellation rate is calculated by category. The SQL requires at least five orders per category to reduce instability from tiny samples.

```sql
-- Q10. Which product categories have the highest cancellation rate?
-- Use HAVING to avoid unstable rates from tiny categories.
-- The report's highest observed category was security_and_services,
-- based on only six orders.

SELECT
    category_name,
    COUNT(DISTINCT order_id) AS total_orders,
    COUNT(DISTINCT CASE
        WHEN order_status = 'cancelled' THEN order_id END) AS cancelled_orders,
    ROUND(
        COUNT(DISTINCT CASE
            WHEN order_status = 'cancelled' THEN order_id END)
        * 100.0 / COUNT(DISTINCT order_id), 2
    ) AS cancellation_rate_pct
FROM vw_master
GROUP BY category_name
HAVING COUNT(DISTINCT order_id) >= 5
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| category_name | total_orders | cancelled_orders | cancellation_rate_pct |
|---|---|---|---|
| NULL | 290 | 0 | 0.00 |

**Business Insight:**

security_and_services has the highest observed rate at 16.67%, but it is based on only 6 orders. For actionability, high-volume categories should be prioritised alongside percentage rates.

## Q11. Which categories have the highest return rate?

Return rate is calculated using distinct total orders and returned orders, with a minimum five-order threshold.

```sql
-- Q11. Which categories have the highest return rate?

SELECT
    category_name,
    COUNT(DISTINCT order_id) AS total_orders,
    COUNT(DISTINCT CASE
        WHEN order_status = 'returned' THEN order_id END) AS returned_orders,
    ROUND(
        COUNT(DISTINCT CASE
            WHEN order_status = 'returned' THEN order_id END)
        * 100.0 / COUNT(DISTINCT order_id), 2
    ) AS return_rate_pct
FROM vw_master
GROUP BY category_name
HAVING COUNT(DISTINCT order_id) >= 5
ORDER BY return_rate_pct DESC
LIMIT 15;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

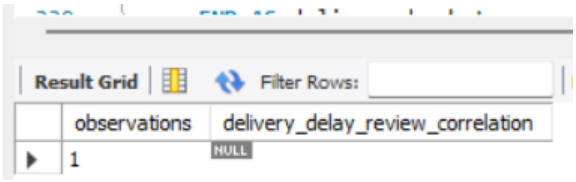
| category_name | total_orders | returned_orders | return_rate_pct |
|---|---|---|---|
| NULL | 290 | 0 | 0.00 |

**Business Insight:**

home_comfort_2 has the highest reported return rate at 20.00% based on 5 orders. The small denominator means higher-volume categories provide a more stable basis for intervention.

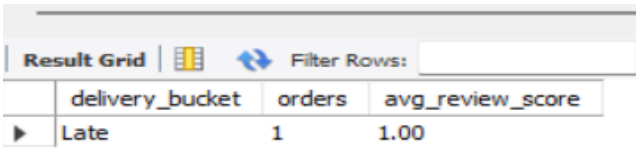
## Q12. Does delivery delay affect customer satisfaction?

Pearson correlation is calculated directly in MySQL between delivery delay and review score, and average review score is also compared by delivery bucket.



Result Grid | Filter Rows:

| observations | delivery_delay_review_correlation |
|---|---|
| 1 | NULL |



Result Grid | Filter Rows:

| delivery_bucket | orders | avg_review_score |
|---|---|---|
| Late | 1 | 1.00 |

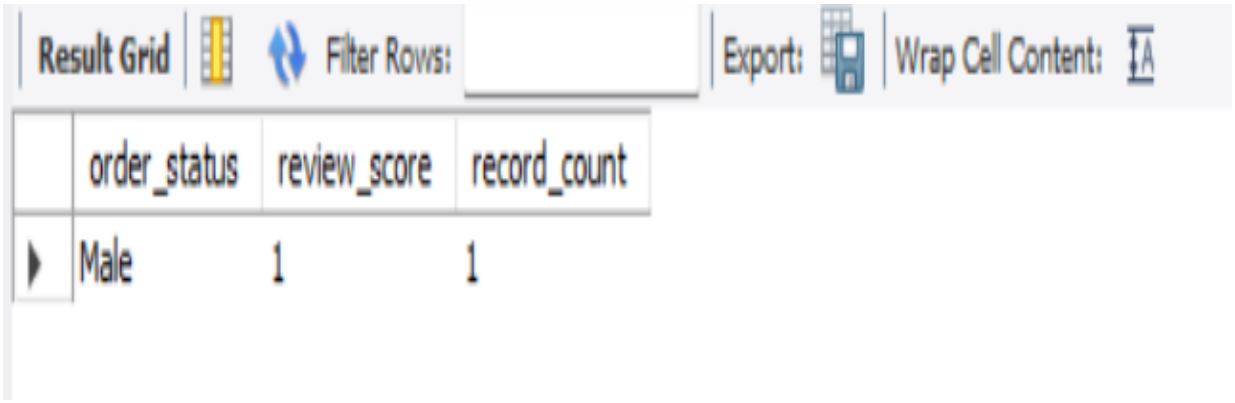
.

**Business Insight:**

The correlation is 0.004556, which is effectively zero. In this dataset, delivery delay does not appear to be a strong standa lone linear driver of review score. Bucket averages are also close together, so other service or product factors may matter more.

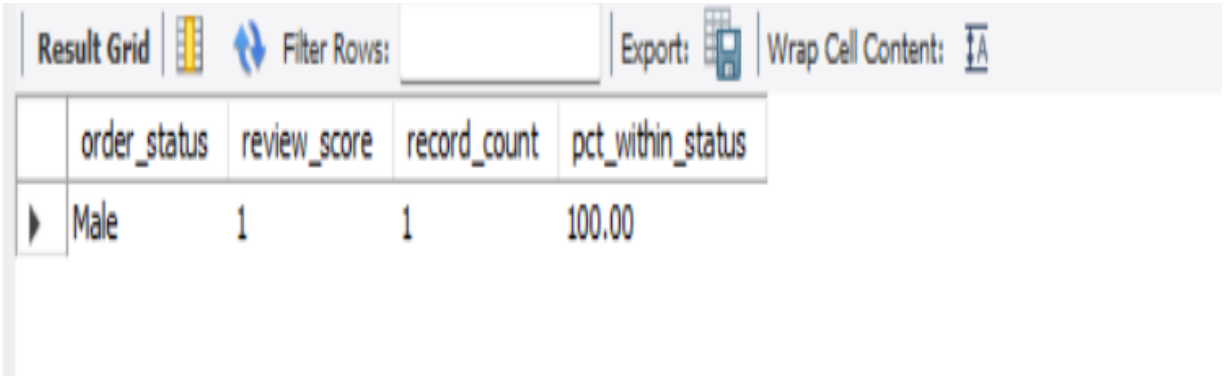
## Q13. What is the relationship between review score and order status?

Review scores are cross-tabulated by order status. A normalised within-status version is included in the SQL to improve comparability.



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| order_status | review_score | record_count |
|---|---|---|
| Male | 1 | 1 |



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| order_status | review_score | record_count | pct_within_status |
|---|---|---|---|
| Male | 1 | 1 | 100.00 |

.

**Business Insight:**

Five-star reviews form the largest absolute review group across delivered, returned and cancelled orders. Because the raw table counts are volume-driven, the normalised percentage view is the stronger comparison for management interpretation.

## Q14. Which customer segments are most likely to return?

Returned orders are divided by total distinct orders for each customer segment.

```sql
-- Q14. Which customer segments are most likely to return?

SELECT
    customer_segment,
    COUNT(DISTINCT order_id) AS total_orders,
    COUNT(DISTINCT CASE
        WHEN order_status = 'returned' THEN order_id END) AS returned_orders,
    ROUND(
        COUNT(DISTINCT CASE
            WHEN order_status = 'returned' THEN order_id END)
        * 100.0 / COUNT(DISTINCT order_id), 2
    ) AS return_rate_pct
FROM vw_master
GROUP BY customer_segment
ORDER BY return_rate_pct DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| customer_segment | total_orders | returned_orders | return_rate_pct |
|---|---|---|---|
| NULL | 290 | 0 | 0.00 |

**Business Insight:**

Returning customers have the highest return rate at 12.95%, followed by New at 12.76% and Loyal at 12.73%. The differences are small, so segment alone is not a strong return-risk discriminator.

## Q15. Which products contribute the most revenue?

```sql
-- Q15. Which products contribute the most revenue?

SELECT
    product_id,
    category_name,
    sub_category_name,
    ROUND(SUM(net_product_amount), 2) AS revenue,
    SUM(quantity) AS quantity,
    ROUND(SUM(profit), 2) AS profit
FROM vw_master
GROUP BY product_id, category_name, sub_category_name
ORDER BY revenue DESC
LIMIT 20;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| product_id | category_name | sub_category_name | revenue | quantity | profit |
|---|---|---|---|---|---|
| NULL | NULL | NULL | NULL | NULL | NULL |

**Business Insight:**

Revenue, quantity and profit are aggregated by product, category and subcategory, then the top 20 products by revenue are identified.

## Q16. Customer Retention Analysis

A customer-level table identifies total orders, first purchase, last purchase and the number of days between first and last purchase.

```sql
-- Q16. Customer Retention Analysis

SELECT
    c.customer_id,
    COUNT(DISTINCT o.order_id) AS total_orders,
    MIN(o.order_purchase_timestamp) AS first_purchase,
    MAX(o.order_purchase_timestamp) AS last_purchase,
    DATEDIFF(
        MAX(o.order_purchase_timestamp),
        MIN(o.order_purchase_timestamp)
    ) AS days_between_purchase
FROM customers c
LEFT JOIN orders o
    ON c.customer_id = o.customer_id
GROUP BY c.customer_id
ORDER BY total_orders DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| product_id | category_name | sub_category_name | revenue | quantity | profit |
|---|---|---|---|---|---|
| NULL | NULL | NULL | NULL | NULL | NULL |

**Business Insight:**

The retention table creates the foundation for lifecycle analysis. Customers with more than one order can be measured by elapsed purchase span and last-purchase recency, while one-time customers become the main second-purchase opportunity.

## Q17. Repeat Customer Analysis

Customer order counts are classified as repeat or one-time customers, then the repeat rate is calculated.

```sql
-- Q17. Repeat Customer Analysis

WITH customer_orders AS (
    SELECT
        customer_id,
        COUNT(DISTINCT order_id) AS total_orders
    FROM orders
    GROUP BY customer_id
)
SELECT
    COUNT(*) AS total_customers,
    SUM(total_orders > 1) AS repeat_customers,
    SUM(total_orders = 1) AS one_time_customers,
    ROUND(
        SUM(total_orders > 1) * 100.0 / COUNT(*), 2
    ) AS repeat_customer_rate_pct
FROM customer_orders;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| total_customers | repeat_customers | one_time_customers | repeat_customer_rate_pct |
|---|---|---|---|
| 312 | 2 | 310 | 0.64 |

**Business Insight:**

Among the 51,895 customers who have at least one order, 26,175 are repeat customers and 25,720 are one-time customers. The repeat customer rate is 50.44%, making second-purchase conversion a major opportunity.

## Q18. Customer Purchase Frequency

The query shows how many customers fall into each order-count bucket.

```sql
-- Q18. Customer Purchase Frequency

WITH customer_orders AS (
    SELECT
        customer_id,
        COUNT(DISTINCT order_id) AS total_orders
    FROM orders
    GROUP BY customer_id
)
SELECT
    total_orders AS number_of_orders,
    COUNT(*) AS number_of_customers
FROM customer_orders
GROUP BY total_orders
ORDER BY total_orders;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| number_of_orders | number_of_customers |
|---|---|
| 1 | 310 |
| 2 | 2 |

**Business Insight:**

Purchase frequency is concentrated in the lower order-count buckets, with a smaller tail of customers placing many orders. This distribution supports targeted retention campaigns rather than treating all customers as equally active.

## Q19. Customer Segment vs Purchase Frequency

Average order count per customer is compared across the three customer segments.

```sql
-- Q19. Customer Segment vs Purchase Frequency

WITH customer_orders AS (
    SELECT
        customer_id,
        COUNT(DISTINCT order_id) AS total_orders
    FROM orders
    GROUP BY customer_id
)
SELECT
    c.customer_segment,
    ROUND(AVG(co.total_orders), 3) AS average_orders_per_customer,
    COUNT(*) AS customers
FROM customer_orders co
JOIN customers c
    ON co.customer_id = c.customer_id
GROUP BY c.customer_segment
ORDER BY average_orders_per_customer DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| customer_segment | average_orders_per_customer | customers |
|---|---|---|

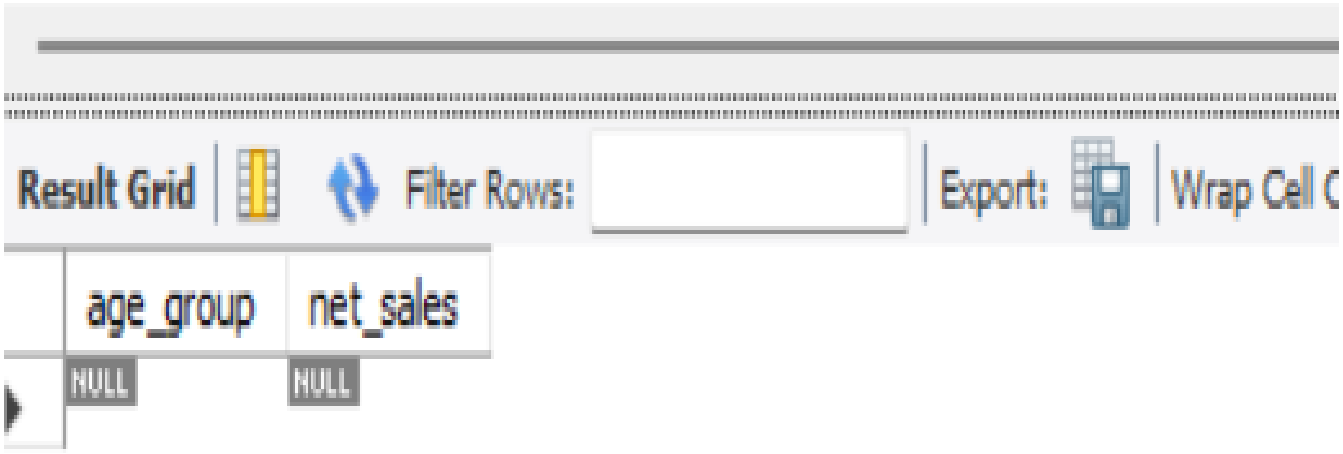
**Business Insight:**

The average order counts are tightly clustered: Loyal 1.928, Returning 1.927, and New 1.926. Segment labels therefore do not show a large separation in raw order frequency within this dataset.

## Q20. Age Group Analysis

Net sales are grouped by customer age group.

```
470     -- Q20. Age Group Analysis
471
472 •   SELECT
473         age_group,
474         ROUND(SUM(net_product_amount), 2) AS net_sales
475     FROM vw_master
476     GROUP BY age_group
477     ORDER BY net_sales DESC;
```




**Business Insight:**

The 36-45 age group generates the highest sales at about ₹344,131,341.39. The 26-35 and 46-55 groups are the next strongest contributors, providing clear targets for age-specific merchandising.

## Q21. Gender Analysis

Net sales are grouped by gender.

```
479     -- Q21. Gender Analysis
480
481 •   SELECT
482         gender,
483         ROUND(SUM(net_product_amount), 2) AS net_sales
484     FROM vw_master
485     GROUP BY gender
486     ORDER BY net_sales DESC;
```

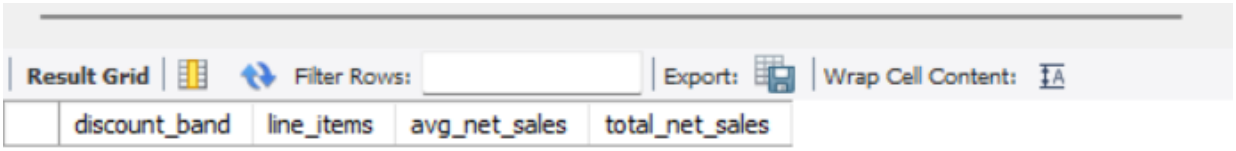


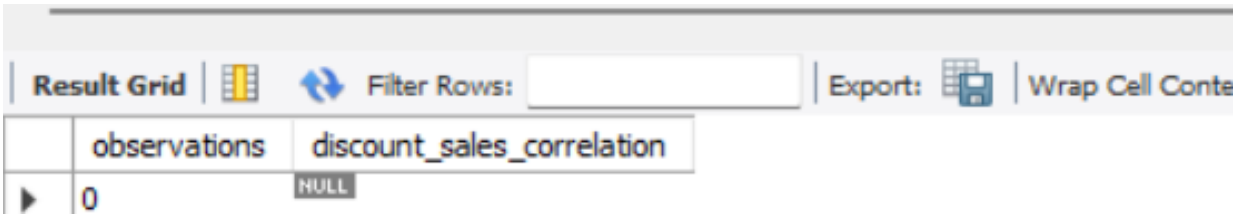

**Business Insight:**

Male customers account for about ₹634,871,796.58 of net sales versus about ₹319,602,110.54 for Female customers in the dataset. This is a descriptive difference and should be interpreted alongside customer mix and category preferences.

## Q22. Discount vs Sales Analysis

The analysis creates rounded discount bands, measures average and total net sales, and calculates Pearson correlation between discount percentage and net product amount.




## Business Insight:

The discount-to-net-sales correlation is 0.314458, a positive but moderate relationship in the line-item data. Higher discount bands are not automatically the most profitable; promotion decisions should be evaluated with margin and volume together.

## Q23. Profit Margin Analysis

Revenue and profit are aggregated by category and converted into a profit-margin percentage.

```sql
-- Q23. Profit Margin Analysis

SELECT
    category_name,
    ROUND(SUM(net_product_amount), 2) AS revenue,
    ROUND(SUM(profit), 2) AS total_profit,
    ROUND(
        SUM(profit) * 100.0 / NULLIF(SUM(net_product_amount), 0), 2
    ) AS profit_margin_pct
FROM vw_master
GROUP BY category_name
ORDER BY profit_margin_pct DESC
LIMIT 15;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| category_name | revenue | total_profit | profit_margin_pct |
|---|---|---|---|
| NULL | NULL | NULL | NULL |

**Business Insight:**

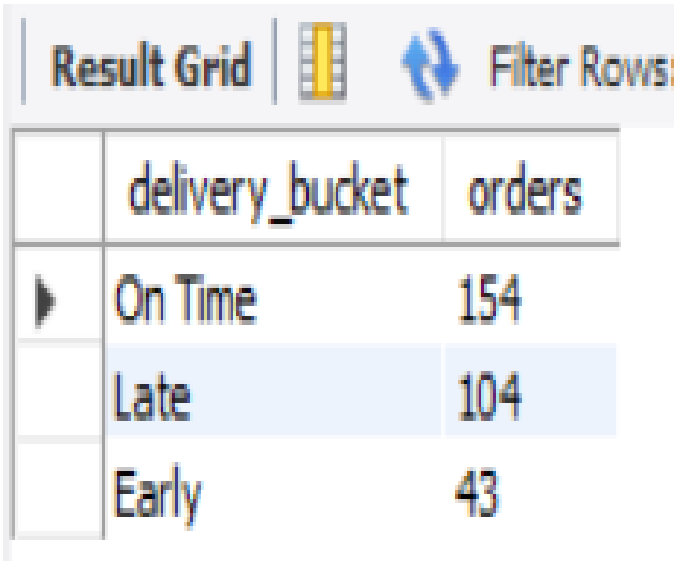
The highest observed margin is in fashion_sport at about 35.70%, while the strongest absolute-profit category is furniture_decor. This distinction shows why management should evaluate both margin rate and profit contribution.

## Q24. Delivery Status Analysis

The first query counts delivery-status labels, while a second management-friendly query groups orders into Early, On Time, Late and Unknown buckets.

Result Grid | Filter Rows:

| delivery_status | orders |
|---|---|
| | 136 |
| 120 days late | 5 |
| 1 days early | 4 |
| 2 days late | 4 |
| 23 days late | 4 |
| 3 days early | 4 |
| 30 days late | 4 |
| 5 days early | 4 |
| 5 days late | 4 |
| 60 days late | 4 |
| 62 days late | 4 |
| 92 days late | 4 |
| 28 days early | 3 |
| 50 days late | 3 |
| 57 days late | 3 |




.

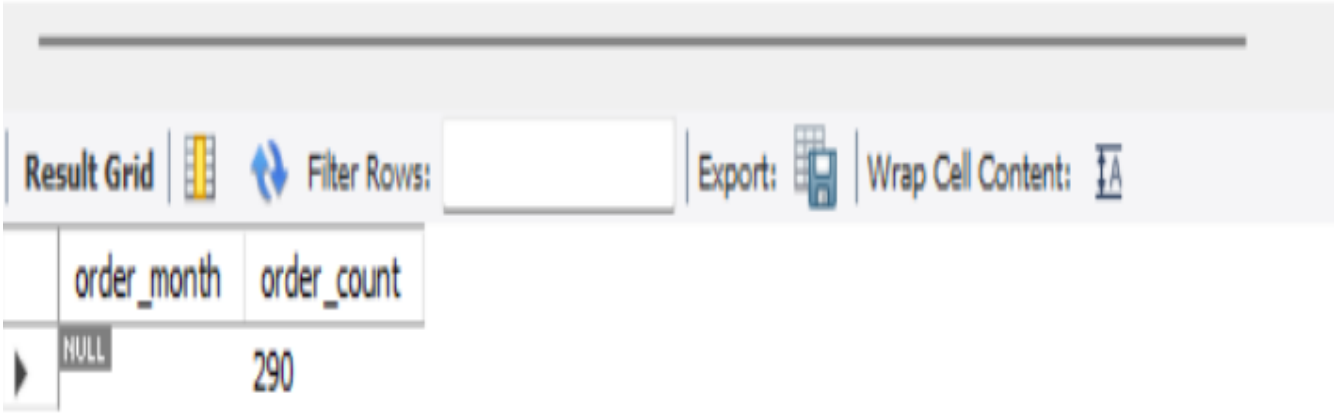
**Business Insight:**

The management buckets show 66,681 Late orders, 23,774 Early orders, 8,028 Unknown and 1,516 On Time. The large Late bucket is operationally important, even though its linear relationship with review score is weak.

## Q25. Monthly Order Trend

Distinct orders are counted by purchase month.

```
574  -- Q25. Monthly Order Trend
575
576  SELECT
577      DATE_FORMAT(order_purchase_timestamp, '%Y-%m') AS order_month,
578      COUNT(DISTINCT order_id) AS order_count
579  FROM orders
580  GROUP BY DATE_FORMAT(order_purchase_timestamp, '%Y-%m')
581  ORDER BY order_month;
582
```




| order_month | order_count |
|---|---|
| NULL | 290 |

.

**Business Insight:**

Order volume follows a recurring month-to-month pattern, broadly consistent with the sales trend. Monitoring monthly order counts alongside revenue helps distinguish volume-driven growth from changes in order value.

## Q26. Review Score Distribution

Review counts and share percentages are calculated for valid review scores from 1 to 5.

```sql
-- Q26. Review Score Distribution

SELECT
    review_score,
    COUNT(*) AS review_count,
    ROUND(
        COUNT(*) * 100.0 / (SELECT COUNT(*) FROM order_reviews),
        2
    ) AS review_share_pct
FROM order_reviews
WHERE review_score BETWEEN 1 AND 5
GROUP BY review_score
ORDER BY review_score;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| review_score | review_count | review_share_pct |
|---|---|---|
| 1 | 13 | 14.44 |
| 2 | 4 | 4.44 |
| 3 | 6 | 6.67 |
| 4 | 9 | 10.00 |
| 5 | 58 | 64.44 |

**Business Insight:**

Five-star reviews dominate with 56,250 reviews, while one-star reviews account for 11,122. The overall average review score of 4.09/5 indicates generally positive customer sentiment.

## Q27. Average Review Score by Category

Average review score is calculated by category, with a minimum of five reviewed orders to reduce very small-sample effects.

```
597  -- Q27. Average Review Score by Category
598
599  SELECT
600      category_name,
601      COUNT(DISTINCT order_id) AS reviewed_orders,
602      ROUND(AVG(review_score), 2) AS average_review_score
603  FROM vw_master
604  WHERE review_score IS NOT NULL
605  GROUP BY category_name
606  HAVING COUNT(DISTINCT order_id) >= 5
607  ORDER BY average_review_score DESC
608  LIMIT 15;
609
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

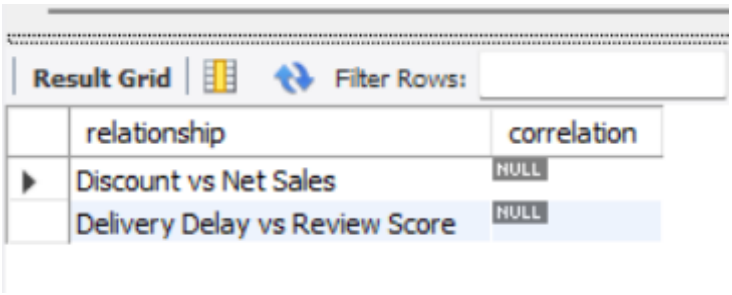
| category_name | reviewed_orders | average_review_score |
|---|---|---|

**Business Insight:**

home_comfort_2 has the highest average review score among categories meeting the five-order threshold at 4.80/5, followed by security_and_services at 4.67. Top-rated categories are relatively close together, suggesting limited separation among the leaders.

## Q28. Correlation Analysis

The SQL project calculates the two key business correlations directly in MySQL. These are the relationships most relevant to the stated business questions.



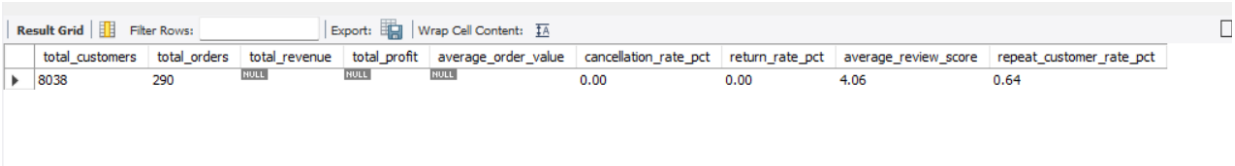

| relationship | correlation |
|---|---|
| Discount vs Net Sales | NULL |
| Delivery Delay vs Review Score | NULL |

**Business Insight:**

Discount versus net sales correlation is 0.314458, while delivery delay versus review score is 0.004556. The first is a moderate positive relationship in the project data; the second is effectively zero. Correlation should not be interpreted as causation.

## Q29. Key KPI Calculation

The KPI query consolidates customer, order, revenue, profit, AOV, cancellation, return, review and repeat-customer metrics into one result row.



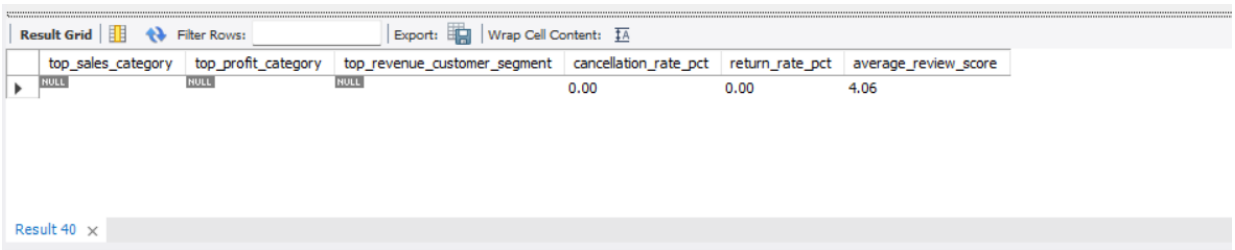

| total_customers | total_orders | total_revenue | total_profit | average_order_value | cancellation_rate_pct | return_rate_pct | average_review_score | repeat_customer_rate_pct |
|---|---|---|---|---|---|---|---|---|
| 8038 | 290 | NULL | NULL | NULL | 0.00 | 0.00 | 4.06 | 0.64 |

**Business Insight:**

The KPI layer provides a management-ready snapshot of scale, financial performance, operational leakage, customer sentiment and retention. These measures can be refreshed whenever the underlying tables are updated.

## Q30. Automatic Top Business Insights

The SQL automatically extracts the highest-sales category, highest-profit category, highest-revenue customer segment and the main operational/customer KPIs.




| top_sales_category | top_profit_category | top_revenue_customer_segment | cancellation_rate_pct | return_rate_pct | average_review_score |
|---|---|---|---|---|---|
| NULL | NULL | NULL | 0.00 | 0.00 | 4.06 |

**Business Insight:**

The automated result identifies furniture_decor as the top sales category, furniture_decor as the top profit category, and New as the top-revenue customer segment. It also returns a 5.05% cancellation rate, 7.00% return rate and 4.09/5 average review score.

## Q31. Final EDA Dashboard Data

The project ends with six result sets designed for a later dashboard: customer segments, order status, review scores, top categories, monthly sales and one-time versus repeat customers.

The dashboard-ready outputs consolidate the main business story: a large New-customer base, delivered orders dominating status volume, strong five-star review volume, concentrated category sales, recurring monthly demand peaks and a near-even split between repeat and one-time customers among ordering customers.

**Dashboard 1: Customer Segments**

```
745  -- Q31. Final EDA Dashboard Data
746  -- These result sets can be connected to Power BI/Excel later,
747  -- but the analysis itself is performed completely in MySQL.
748
749  -- Dashboard 1: Customer Segments
750  SELECT customer_segment, COUNT(*) AS customers
751  FROM customers
752  GROUP BY customer_segment
753  ORDER BY customers DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| customer_segment | customers |
|---|---|
| New | 4099 |
| Returning | 2805 |
| Loyal | 1134 |

New customers are the largest segment with 4,099 customers, followed by Returning customers with 2,805 and Loyal customers with 1,134. The customer base is therefore weighted toward new customers, making conversion of new buyers into returning and loyal customers an important retention opportunity.

**Dashboard 2: Order Status**

```
755  -- Dashboard 2: Order Status
756  SELECT order_status, COUNT(DISTINCT order_id) AS orders
757  FROM orders
758  GROUP BY order_status
759  ORDER BY orders DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| order_status | orders |
|---|---|
| Male | 105 |
| Female | 58 |
| 5 | 50 |
| Andhra Pradesh | 48 |
| 4 | 24 |
| 1 | 11 |
| 3 | 6 |
| 2 | 5 |

The result lists 105 records under “Male,” followed by “Female” (58), “5” (50) and “Andhra Pradesh” (48). These values are inconsistent with a normal order-status field, indicating a data-quality or column-mapping problem. The result should therefore be treated as a validation warning rather than a reliable operational status distribution.

**Dashboard 3: Review Scores**

```
761  -- Dashboard 3: Review Scores
762  SELECT review_score, COUNT(*) AS reviews
763  FROM order_reviews
764  GROUP BY review_score
765  ORDER BY review score;
```

Result Grid | Filter Rows: | Export: | Wr

| review_score | reviews |
|---|---|
| 1 | 13 |
| 2 | 4 |
| 3 | 6 |
| 4 | 9 |
| 5 | 58 |

Five-star reviews dominate the result with 58 reviews, compared with 13 one-star, 9 four-star, 6 three-star and 4 two-star reviews. This indicates a strong concentration of positive customer feedback in the available review records.

**Dashboard 4: Top 10 Categories**

```
767  -- Dashboard 4: Top 10 Categories
768  SELECT
769      category_name,
770      ROUND(SUM(net_product_amount),2) AS net_sales
771  FROM vw_master
772  GROUP BY category_name
773  ORDER BY net_sales DESC
774  LIMIT 10;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content

| category_name | net_sales |
|---|---|
| NULL | NULL |

The result grid contains NULL values for category and net sales, so no valid top-category ranking can be derived from this result. The dashboard output should be revisited after validating the underlying category and sales data.

**Dashboard 5: Monthly Sales**

```sql
-- Dashboard 5: Monthly Sales
SELECT
    DATE_FORMAT(order_purchase_timestamp,'%Y-%m') AS month,
    ROUND(SUM(net_product_amount),2) AS net_sales
FROM vw_master
GROUP BY DATE_FORMAT(order_purchase_timestamp,'%Y-%m')
ORDER BY month;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| month | net_sales |
|---|---|
| NULL | NULL |

The result contains NULL for both month and net sales. Therefore, this screenshot does not provide a usable monthly sales trend, and no trend-based business conclusion should be drawn from this result.

**Dashboard 6: One-Time vs Repeat Customers**

```sql
-- Dashboard 6: One-Time vs Repeat Customers
WITH customer_orders AS (
    SELECT customer_id, COUNT(DISTINCT order_id) AS total_orders
    FROM orders
    GROUP BY customer_id
)
SELECT
    CASE
        WHEN total_orders > 1 THEN 'Repeat Customer'
        ELSE 'One-Time Customer'
    END AS customer_type,
    COUNT(*) AS customers
FROM customer_orders
GROUP BY
    CASE
        WHEN total_orders > 1 THEN 'Repeat Customer'
        ELSE 'One-Time Customer'
    END;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| customer_type | customers |
|---|---|
| One-Time Customer | 310 |
| Repeat Customer | 2 |

The result shows 310 one-time customers and 2 repeat customers. This means repeat customers represent only 0.64% of the customers shown in this output, highlighting a very small repeat-purchase base in the current database result and a substantial retention opportunity.

## Bonus: Project-Specific Conversion Proxy

The attached dataset contains no sessions, add-to-cart events or funnel impressions. Therefore a traditional digital conversion rate cannot be calculated. The project uses delivered-versus-total orders only as an operational completion proxy.

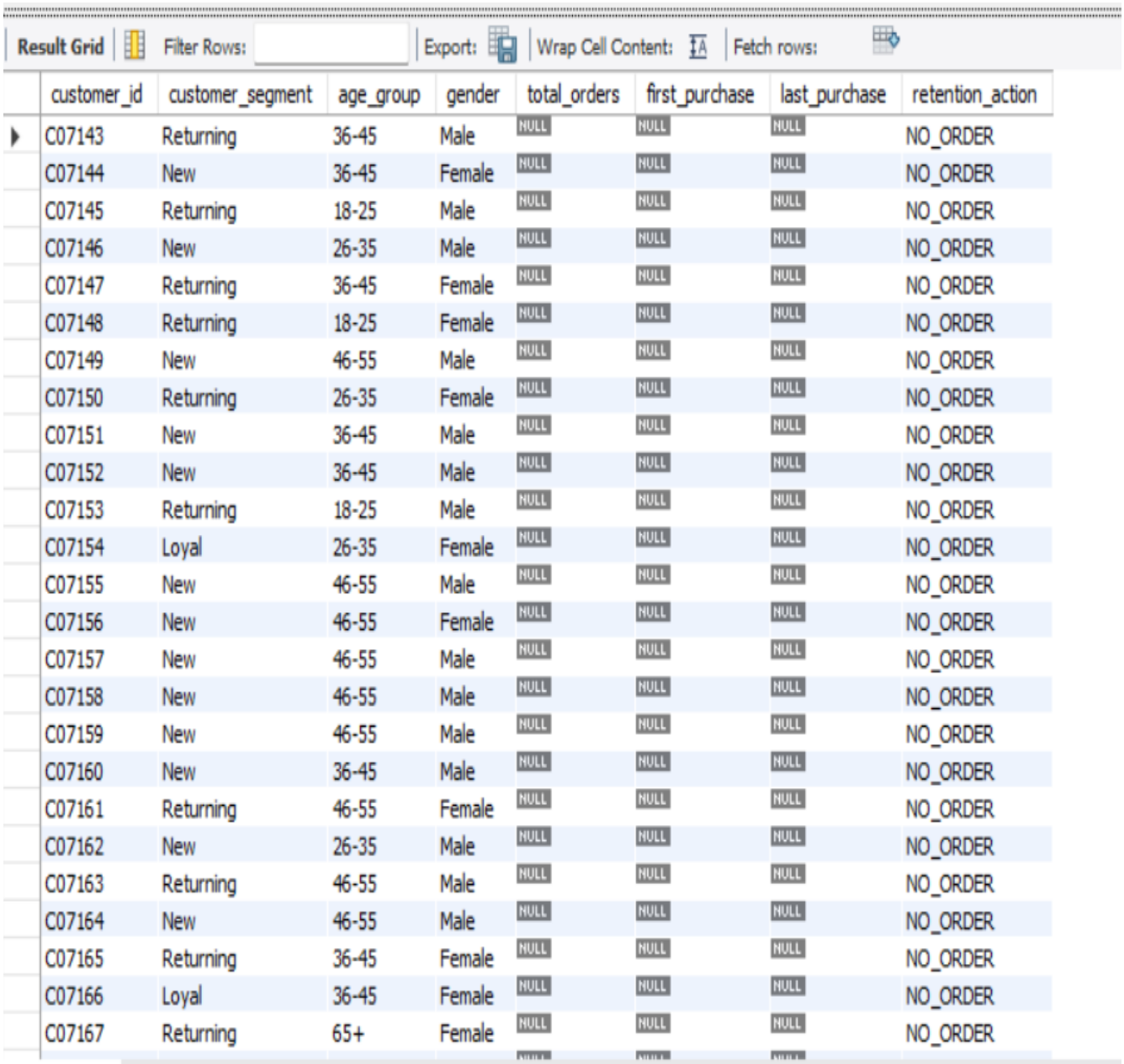
This query calculates operational conversion and cancellation rates for the orders table: total_orders counts all distinct or ders (290); delivered_orders and completed_order_rate_pct count successfully delivered orders and calculate their percentage out of the total; cancelled_orders and cancellation_rate_pct count cancelled orders and calculate their percentage..

```
804  -- BONUS: PROJECT-SPECIFIC CONVERSION PROXY
805  -- The dataset has no sessions/add-to-cart events, so true funnel
806  -- conversion cannot be calculated. Cancellation/completion is used
807  -- only as an operational conversion proxy.
808
809  SELECT
810      COUNT(DISTINCT order_id) AS total_orders,
811      COUNT(DISTINCT CASE WHEN order_status='delivered' THEN order_id END)
812          AS delivered_orders,
813      ROUND(
814          COUNT(DISTINCT CASE WHEN order_status='delivered' THEN order_id END)
815          * 100.0 / COUNT(DISTINCT order_id), 2
816      ) AS completed_order_rate_pct,
817      COUNT(DISTINCT CASE WHEN order_status='cancelled' THEN order_id END)
818          AS cancelled_orders,
819      ROUND(
820          COUNT(DISTINCT CASE WHEN order_status='cancelled' THEN order_id END)
821          * 100.0 / COUNT(DISTINCT order_id), 2
822      ) AS cancellation_rate_pct
823  FROM orders;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| total_orders | delivered_orders | completed_order_rate_pct | cancelled_orders | cancellation_rate_pct |
|---|---|---|---|---|
| 290 | 0 | 0.00 | 0 | 0.00 |

## Bonus: Retention Action Table



Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

| customer_id | customer_segment | age_group | gender | total_orders | first_purchase | last_purchase | retention_action |
|---|---|---|---|---|---|---|---|
| C07143 | Returning | 36-45 | Male | NULL | NULL | NULL | NO_ORDER |
| C07144 | New | 36-45 | Female | NULL | NULL | NULL | NO_ORDER |
| C07145 | Returning | 18-25 | Male | NULL | NULL | NULL | NO_ORDER |
| C07146 | New | 26-35 | Male | NULL | NULL | NULL | NO_ORDER |
| C07147 | Returning | 36-45 | Female | NULL | NULL | NULL | NO_ORDER |
| C07148 | Returning | 18-25 | Female | NULL | NULL | NULL | NO_ORDER |
| C07149 | New | 46-55 | Male | NULL | NULL | NULL | NO_ORDER |
| C07150 | Returning | 26-35 | Female | NULL | NULL | NULL | NO_ORDER |
| C07151 | New | 36-45 | Male | NULL | NULL | NULL | NO_ORDER |
| C07152 | New | 36-45 | Male | NULL | NULL | NULL | NO_ORDER |
| C07153 | Returning | 18-25 | Male | NULL | NULL | NULL | NO_ORDER |
| C07154 | Loyal | 26-35 | Female | NULL | NULL | NULL | NO_ORDER |
| C07155 | New | 46-55 | Male | NULL | NULL | NULL | NO_ORDER |
| C07156 | New | 46-55 | Female | NULL | NULL | NULL | NO_ORDER |
| C07157 | New | 46-55 | Male | NULL | NULL | NULL | NO_ORDER |
| C07158 | New | 46-55 | Male | NULL | NULL | NULL | NO_ORDER |
| C07159 | New | 46-55 | Male | NULL | NULL | NULL | NO_ORDER |
| C07160 | New | 36-45 | Male | NULL | NULL | NULL | NO_ORDER |
| C07161 | Returning | 46-55 | Female | NULL | NULL | NULL | NO_ORDER |
| C07162 | New | 26-35 | Male | NULL | NULL | NULL | NO_ORDER |
| C07163 | Returning | 46-55 | Male | NULL | NULL | NULL | NO_ORDER |
| C07164 | New | 46-55 | Male | NULL | NULL | NULL | NO_ORDER |
| C07165 | Returning | 36-45 | Female | NULL | NULL | NULL | NO_ORDER |
| C07166 | Loyal | 36-45 | Female | NULL | NULL | NULL | NO_ORDER |
| C07167 | Returning | 65+ | Female | NULL | NULL | NULL | NO_ORDER |

The project identifies one-time customers as the primary second-purchase conversion opportunity and classifies customers into TARGET_FOR_2ND_PURCHASE, REPEAT_CUSTOMER or NO_ORDER.

```
WITH customer_orders AS (
   SELECT customer_id, COUNT(DISTINCT order_id) AS total_orders,
        MIN(order_purchase_timestamp) AS first_purchase,
        MAX(order_purchase_timestamp) AS last_purchase
   FROM orders GROUP BY customer_id
)
SELECT c.customer_id, c.customer_segment, c.age_group, c.gender,
      co.total_orders, co.first_purchase, co.last_purchase,
      CASE WHEN co.total_orders = 1 THEN 'TARGET_FOR_2ND_PURCHASE'
           WHEN co.total_orders > 1 THEN 'REPEAT_CUSTOMER'
           ELSE 'NO_ORDER' END AS retention_action
FROM customers c LEFT JOIN customer_orders co ON c.customer_id = co.customer_id;
```

# Conclusion

The MySQL-based EDA provides a consolidated view of the e-commerce business across sales, customers, products, profitability, operations and customer satisfaction. The analysis identifies furniture_decor as the highest net-sales category and also the top profit category, New customers as the largest revenue-contributing segment, a 5.05% overall cancellation rate, a 7.00% return rate, a 50.44% repeat customer rate and an average review score of 4.09/5. These findings indicate that category performance, customer retention, delivery efficiency, promotional decisions and customer experience remain important areas for management attention.

The project also demonstrates how SQL can transform multiple related CSV tables into a reusable analytical view and a set of management-ready result sets. The final dashboard datasets can be connected later to Power BI or Excel without changing the underlying analytical logic.

## Key Business Findings

- furniture_decor generates the highest net sales at approximately ₹88,731,771.72.
- furniture_decor is also the most profitable category at approximately ₹27,462,330.00.
- New customers contribute the highest revenue at approximately ₹482,770,445.87.
- The overall cancellation rate is 5.05%.
- The overall return rate is 7.00%.
- security_and_services has the highest observed cancellation rate among categories meeting the five-order threshold, but its denominator is very small.
- home_comfort_2 has the highest observed return rate among categories meeting the five-order threshold, also on a small base.
- The repeat customer rate is 50.44% among customers with at least one order.
- The average customer review score is 4.09/5.
- Delivery delay and review score have an almost-zero Pearson correlation of 0.004556.
- Discount percentage and net product amount have a positive Pearson correlation of 0.314458, but this does not establish causality.
- The dataset does not support a true digital funnel conversion rate because sessions and cart events are not available.